

Effect of velocity distribution on red cell distribution in capillary blood vessels

R. T. YEN AND Y. C. FUNG

*Department of AMES-Bioengineering, University of California, San Diego,
La Jolla, California 92093*

YEN, R. T., AND Y. C. FUNG. *Effect of velocity distribution on red cell distribution in capillary blood vessels.* Am. J. Physiol. 235(2): H251-H257, 1978 or Am. J. Physiol.: Heart Circ. Physiol. 4(2): H251-H257, 1978.—Through the use of simulated model experiments, data on blood cell distribution into a bifurcating capillary blood vessel are obtained. The results show that the movement of red blood cells at a bifurcation point is influenced by the difference in velocities of flow in the daughter branches. If the velocity of flow in one branch is slower than that in the other, the hematocrit decreases in the slower branch and increases in the faster branch. For velocity ratios sufficiently smaller than a certain critical value, the hematocrit ratio can be expressed by a linear relationship, $(H_1/H_2) - 1 = a|(v_1/v_2) - 1|$, in which v_1 , v_2 and H_1 , H_2 denote the particle velocities and tube hematocrits in the branches 1 and 2, respectively, and a is a dimensionless constant dependent upon a number of factors, the most important of which are the ratio of cell diameter to tube diameter, the shape and rigidity of the pellets, and the hematocrit in the feeding tube. For velocity ratios beyond a critical value, nearly all the cells flow into the faster branch. The smaller the feeding tube hematocrit is, the smaller is the critical velocity ratio at which this phenomenon occurs.

model experiment; critical velocity ratio; hematocrit: tube, feed-tube, discharge; velocity: particle, mean flow

SINCE OXYGEN and CO₂ exchange takes place in red blood cells, the distribution of red blood cells in capillary blood vessels has physiological importance. It is therefore important to know how red cells distribute themselves in the microvascular bed. In this report we shall show that the red cell distribution in capillary blood vessels depends on the velocity distribution. The faster channels get more red cells. On the other hand, the velocity of flow in every vessel is directly proportional to the pressure gradient and inversely proportional to the apparent viscosity, whereas the apparent viscosity is increased by increased red cell concentration. Hence in any specific circuit (vessel system) there is a kind of feedback and modulation which determines the flow, pressure, and red cell distributions. The red cell distribution-velocity relationship is one link in this system.

Its basic cause is biophysical. It does not depend on, although it is superposed to, the active control of smooth muscle action.

To clarify the meaning of the velocity-hematocrit relationship, let us consider a narrow capillary blood vessel with a diameter comparable to that of the red blood cells. Let this blood vessel bifurcate into two daughter branches at a certain point. At the bifurcation point, a red blood cell flowing down the capillary must flow into one of the two daughter branches. The forces that determine into which branch the red cell must go are the resultants of the pressure and the shear stress in the fluid. Both of these resultants tend to pull the cell into the faster channel. In the simplest case of rigid spherical balls flowing down a circular cylindrical tube that branches, this can be analyzed and demonstrated very easily. Using balls of diameter about 90% of the diameter of the tube, Fung (1) has shown that the faster branch gets all the balls, the slower branch gets none. The outcome is less certain if the ratio of ball diameter to tube diameter is smaller or if the balls are not spherical, or if they are deformable rather than rigid. Red blood cells are not spherical and are very flexible. Hence they demand a more careful analysis. The purpose of this article is to supply some data to answer the question of red cell distribution into bifurcating capillary blood vessels.

Two types of experiments may be done to resolve the problem quantitatively: *a*) model experiments and *b*) observation in vivo. A model experiment is simple and accurate, but is not real microcirculation. It can be used to demonstrate a principle and to obtain accurate data to assess theoretical calculations. In vivo observations in some animals would be desirable, but there are difficulties. 1) A very accurate method of determining the blood vessel diameter in vivo is not available. 2) The whole blood contains white blood cells which, though much fewer in number than red blood cells, do exercise considerable influence on the red cell distribution, because some of them are much bigger than the red cells and are almost spherical in shape and hence would move into the faster branch much more readily

These white cells often carry a string of closely packed red cells behind them. 3) The only method available for measuring the velocity of flow in capillaries is by measuring the velocity of red cells. Since the velocity in a capillary vessel is not uniform, certain corrections are needed. These difficulties are not so serious as to completely invalidate *in vivo* observations, but they do complicate the problem.

A model approach is adopted for the present study. In our model, the plasma is simulated by silicone fluid, the red blood cells are simulated by gelatin pellets, and the blood vessel is simulated by Lucite tubes. The dimensions of the model are designed according to the principle of kinematic and dynamic similarity of flow. The entire model is large enough to permit visual observation of flow details. Instrumentation is simple and measurements are quite accurate.

The results of the model experiment show that the movement of red cells at the bifurcation point is influenced by the velocities of flow in the branches. If the velocity of flow in one capillary branch is slower than that in the other, the hematocrit decreases in the slower branch and increases in the faster branch. Within a certain limit of the ratio of the velocities in the two branches, the ratio of the hematocrits in the branches is proportional to the velocity ratio. Beyond a certain velocity ratio the hematocrit ratio increases rapidly in such a way as to suggest that the faster branch gets all the red cells. That limiting value will be called the critical velocity ratio. The transition from the linear regime into the critical condition is quite fast. It follows that if the velocity ratio is sufficiently smaller than the critical value, and if the two branches have the same cross-sectional area, then the ratio of the volume flow rates in the branches is proportional to the ratio of the hematocrits in the branches. In this case the ratio of the red cell flow (volume of red cells/s) in these branches is proportional to the ratio of the squares of the hematocrits in these branches. For velocity ratios greater than the critical value, however, the hematocrit ratio will no longer be indicative of relative velocities of flow in the two branches.

METHOD

Simulated red blood cells—gelatin pellets. A suspension of gelatin pellets in silicone fluid was used to simulate the blood. The pellets were circular cylindrical disks with a diameter of 3.2 mm and a thickness of 1 mm. Since the experiments require large numbers of cells, a cell-cutting device was fabricated. This device is a piston-and-cylinder arrangement with long metal tubes (with uniform inner diameter of 0.32 cm) acting as cylinders. The tubes are filled with a liquid gelatin which hardens upon cooling. The gelatin in the cylinder is pushed out by a piston at a desired speed. As the gelatin is pushed out, it is sliced by a fine wire oscillating up and down by a rotating eccentric wheel. The ejection of the gelatin and hence the thickness of the cell is accurately controlled by a digital computer (PDP8E) that switches on a stepping motor. In order to closely match the specific gravity of the gelatin pellets

and silicone fluid (specific gravity, 0.97), a suitable amount of alcohol (isopropyl) is added to the water when the gelatin pellets are made. The composition and the elastic properties of the gelatin pellets were reported earlier (8). The experiments were carried out with flows in the Reynolds range 10^{-2} – 10^{-4} , which thus simulates the range of Reynolds numbers found in microcirculation *in vivo*. Dow-Corning silicone fluid with a kinematic viscosity of 100 stokes was used for simulation of plasma.

Apparatus and experimental procedure. A schematic diagram of the experimental apparatus is given in Fig. 1. The apparatus consists of a closed reservoir of simulated blood with an inverted T tube of Lucite attached as shown. The tube simulates a bifurcating capillary blood vessel. All branches are circular cylinders and of the same diameter. A shut-off valve is positioned between the reservoir and the testing tube to stop the flow when desired. In the experiments, three different sizes of Lucite testing tubes were used. A source of constant pressure, connected to the top of the reservoir, drove the fluid through the test tube. The ends of the tubes, as well as the junction of the T (the bifurcation point), were carefully machined. The radius of curvature of the edge of junction (the corner of the branches at bifurcation) was measured in a comparator after cutting open a test tube. The radius of curvature was found to be 0.03–0.05 mm, or about 1–1.5% of the simulated red cell diameter, 3–5% of the cell thickness. A motor-driven stirrer kept the suspension uniformly mixed during the experiment.

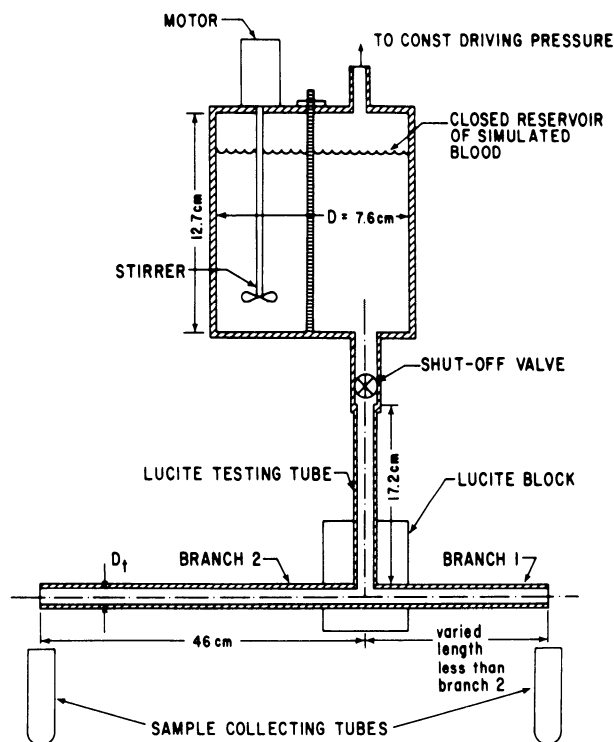


FIG. 1. Schematic diagram of test model of red blood cell flow in a bifurcating capillary blood vessel. Three cylindrical tubes meet to form an inverted T joint. Flow in branch 1 is faster than in branch 2. Inner diameters of test tube D_1 are 0.32, 0.48, and 0.64 cm.

The hematocrit and velocity relationship in bifurcation branches was determined by allowing a gelatin pellet suspension to flow through the T tube. The experiment consisted of collecting quantities of fluid from the two branches in a steady-state condition. Flow can be stopped by closing the shut-off valve after each test. Velocities in the two branches were varied by changing the lengths of the branches by cutting off a segment of one branch at a time. In each experiment, the suspension of pellets in the reservoir was carefully mixed and stirred to a homogeneous concentration. The driving pressure was kept at a constant value. The pellet velocities were obtained by measuring the time required for a pellet to travel a known distance of 8.9 cm. The mean velocity of flow in the tube was computed from the volume discharge collected in a known interval of time. The pellet velocities and the mean velocities of flow were measured simultaneously in two tubes (branch 1 and branch 2). Four stop watches were used for the velocity measurements by two persons. To determine the discharge hematocrit, the collected sample from each branch was centrifuged for 1 h at $3,000 \times g$; the packed particle volume and total volume were measured, and their ratio was computed. Although the specific gravities of gelatin pellets and silicone fluid were originally closely matched, by the time of centrifugation enough alcohol had evaporated so as to make the resultant specific gravities slightly different and centrifugation possible. When using this centrifugation technique, we found that the silicone fluid trapped in the packed pellets may amount to 3% of the pellet volume (9). Corrections of the data were made accordingly.

DATA ANALYSIS

The hematocrit in each tube was determined from their respective discharge hematocrit using the following relationship (6)

$$\frac{\text{discharge hematocrit}}{\text{tube hematocrit}} = \frac{\text{mean speed of pellets}}{\text{mean speed of tube flow}} \quad (1)$$

Hematocrit (H_f) of the feeding tube was calculated as follows. Let $\bar{V}_f$ be the mean flow velocity in the feeding tube, $\bar{V}_1$ be that in branch 1, $\bar{V}_2$ be that in branch 2; and let v_f be the particle velocity in the feeding tube, v_1 be that in branch 1, v_2 be that in branch 2. Without loss of generality, we may let $v_1 \geq v_2$, and $\bar{V}_1 \geq \bar{V}_2$.

Then, by the principle of conservation of mass of the mixtures and of the pellets, we have, since the cross-sectional area of all branches are the same,

$$\bar{V}_f = \bar{V}_1 + \bar{V}_2 \quad (2)$$

$$H_f \bar{V}_f = H_1 \bar{V}_1 + H_2 \bar{V}_2 \quad (3)$$

The mean flow velocities $\bar{V}_f$, $\bar{V}_1$, $\bar{V}_2$ are related to the particle velocities v_f , v_1 , v_2 . These velocities were measured and the ratios $v_1/\bar{V}_1$, $v_2/\bar{V}_2$ were computed. The results are shown in Table 1 and Fig. 5. It is seen that the ratio of particle velocity to mean flow velocity depends strongly on the ratio of cell diameter to tube diameter, D_c/D_t , but does not vary significantly with

hematocrit. Nor does it vary with flow rate. Hence for each value of D_c/D_t , we can replace Eq. 2 by

$$v_f = v_1 + v_2 \quad (4)$$

On substituting Eq. 4 into Eq. 3, we obtain the feed tube hematocrit H_f

$$H_f = \frac{H_1 v_1 + H_2 v_2}{v_1 + v_2} \quad (5)$$

RESULTS

Figures 2-4 show the experimental results. In these figures, the ratio H_1/H_2 is plotted against the ratio v_1/v_2 for different feed-tube hematocrits H_f . The symbols v_1 , v_2 , H_1 , H_2 are the particle velocities and tube hematocrits in the branches 1 and 2 respectively. It is seen that for narrow capillaries (with the cell diameter/tube diameter = $D_c/D_t = 1.0, 0.67$, and 0.5) the branch with faster flow (branch 1) will have more cells. In general, for velocity ratios sufficiently smaller than a critical value, the hematocrit ratio can be expressed by a linear relationship given by

$$\frac{H_1}{H_2} - 1 = a \left(\frac{v_1}{v_2} - 1 \right) \quad (6)$$

In Eq. 6, the dimensionless constant, a , depends on a

TABLE 1. Ratio of particle velocity to mean flow velocity

D_c/D_t	Hematocrit			Combined Data
	10%	20%	30%	
1.0	1.21 ± 0.012	1.17 ± 0.012	1.20 ± 0.016	1.19 ± 0.008
0.67	1.47 ± 0.010	1.42 ± 0.014		1.45 ± 0.008
0.5	1.49 ± 0.014	1.45 ± 0.014		1.48 ± 0.011

Experimental results of test model for $D_c/D_t = 1.0, 0.67$, and 0.5 . Means $\pm$ SE of the ratio between particle velocity and mean flow velocity are listed corresponding to feed-tube hematocrit and combined for each value of D_c/D_t .

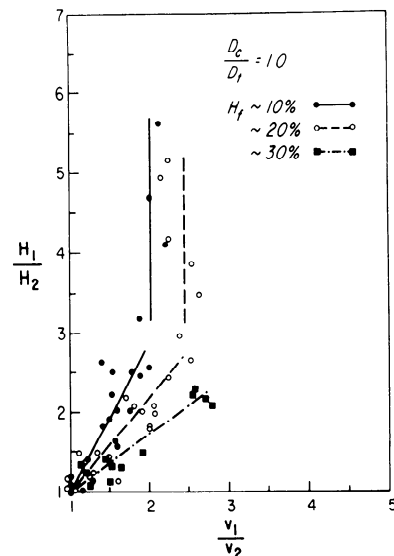


FIG. 2. Experimental results of test model for $D_c/D_t = 1.0$. H_1/H_2 is plotted against v_1/v_2 for three different feed-tube hematocrits H_f .

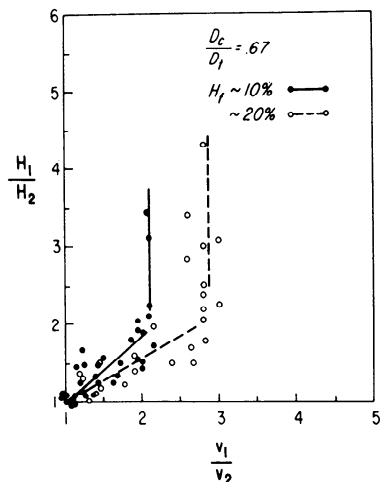


FIG. 3. Experimental results of test model for $D_c/D_t = 0.67$. H_1/H_2 is plotted against v_1/v_2 for two different feed-tube hematocrits H_f .

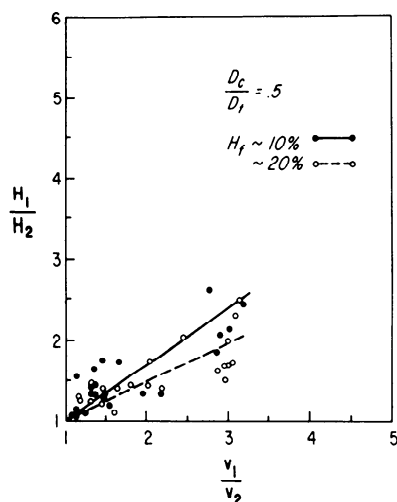


FIG. 4. Experimental results of test model for $D_c/D_t = 0.5$. H_1/H_2 is plotted against v_1/v_2 for two different feed-tube hematocrits H_f .

number of factors, the most important of which are *i*) the ratio of cell diameter to tube diameter, *ii*) the shape and rigidity of the pellets, and *iii*) the hematocrit in the feeding tube.

Figs. 2 and 3 also show that for velocity ratios beyond a critical value nearly all the cells flow into the faster branch. The smaller the feeding tube hematocrit is, the smaller is the critical velocity ratio at which this phenomenon occurred. The critical velocity ratio lies in the range of 2–3.0 when $D_c/D_t = 1.0$, with exact value depending on the feed hematocrit. The critical velocity ratio becomes higher when D_c/D_t decreases.

In Fig. 5 the ratio of particle velocity to mean flow velocity is plotted against the particle velocity for different values of D_c/D_t and hematocrit. The dotted line in the figure represents the mean value of all data for each value of D_c/D_t . It is seen that the particle-to-mean velocity ratio is independent of the flow rate, but dependent on D_c/D_t .

Table 1 lists the mean values and standard errors of the particle-to-mean velocity ratio at hematocrits 10, 20, and 30%. The last column gives the velocity ratio of all pooled data.

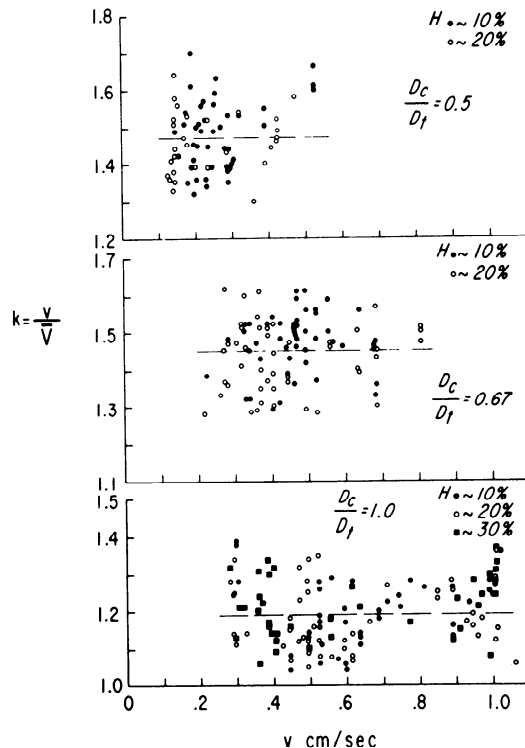


FIG. 5. Experimental results of test model for $D_c/D_t = 0.5, 0.67$, and 1.0 . The ratio k between particle velocity v and mean flow velocity $\bar{V}$ is plotted against v for different feed-tube hematocrits H_f . Dotted lines are mean values of k computed for combined values of H_f for each value of D_c/D_t .

DISCUSSION

Blood flow in capillary blood vessels is often seen to be unsteady. At one instant, one may see a capillary without any red blood cell; in the next instant he may see the same capillary packed with flowing red cells. The capillaries without red cells are often thought to be closed by precapillary sphincters, but the result presented above shows that this interpretation is not infallible. If the velocity ratio in two branches exceeds the critical value, (of the order of 2–4), then the slower branch may very well have a flow of plasma but with very few red cells. In other words, a very uneven distribution of red cells can be created by the velocity difference in the branches, even without the contraction of sphincters. This phenomenon was observed by Svanes and Zweifach (7) by means of a microocclusion technique. They cleared a capillary vessel of its blood cells by occluding the downstream end of the vessel, which effectively altered the velocity ratio in capillary branches.

The situation is similar in pulmonary circulation. The role of velocity ratio is clearer in this case because these pulmonary capillaries do not have precapillary sphincters, yet the patchy filling of pulmonary capillary bed is well known.

In our experiment, we chose a model with a 180° angle of bifurcation. This choice of angle was arbitrary. Perhaps such an angle of bifurcation is not common in microcirculation. But we believe changes in angle between bifurcating branches will not affect the observed phenomenon, because the factors which cause a red cell

to take the branch with a faster flow are virtually independent of the angle. The factors are the net pressure force and the net shear force acting on the cell. The pressure force depends on the pressure gradient which is larger on the side with faster flow. The shear force varies linearly with shear gradient, which is again larger on the faster side. The angle of branching would have only a relatively minor effect. A more detailed discussion of this point is given in the APPENDIX.

The sharp edge of the bifurcation joint was used in order to eliminate the radius of curvature of the edge as another variable in this investigation. The Reynolds number of flow was so low that the flow was laminar at all times. Flow separation was not observed at the corner.

The pellet's geometry and deformability also affects the distribution of the cells at the point of bifurcation. Comparison of the present results on disklike flexible gelatin pellets with Fung's (1) rigid ball experiments clearly demonstrates the same tendency of suspended particles to go to the faster side in both cases. But the flexible, disklike gelatin pellets require a larger difference in velocities between the two branches for the faster branch to get all the particles. Therefore, if we are permitted to extrapolate the results, we may say that the more flexible the particles are, the more likely it is that the hematocrit will be proportional to the velocity of flow in branches and the higher will be the critical velocity ratio. A semiquantitative analysis of this point is given in the APPENDIX.

Model experiments by Lee and Fung (4) in which a single cell was modeled to move in a capillary vessel suggest a linear relationship between pellet velocity and mean flow velocity. They observed an increase in the particle-to-mean velocity ratio as the ratio D_c/D_t decreased. Our data shows a similar trend. The mean value of the particle-to-mean velocity ratio increases from 1.19 to 1.48 as D_c/D_t decreases from 1.0 to 0.5. The dispersion of data in Fig. 5 can be attributed to some nonuniformity of pellet size resulting from broken cells and some cells adhering together. It is seen from the figure that the velocity ratio is not significantly affected by hematocrit and flow rate in the range studied here.

The result of the present experiment also has an entirely different application. It is quite popular today to use radioactive microspheres in blood flow to measure the distribution of blood flow in different parts of organs. Now the distribution of microspheres into bifurcating vessels of relevant sizes must be subjected to exactly the same kind of fluid mechanical forces as the red blood cells in capillaries. Hence microspheres of 50 μm diameter will be controlled by the velocity distribution in blood vessels of diameters in the neighborhood of 50–100 μm . If the measurement of radiation is proportional to the total number of microspheres per unit volume, then it will indicate the distribution of blood flow only if the numerical concentration of microspheres per unit volume of blood is uniform; and this happens only if the velocity of flow at points of bifurcation into different branches is equal. If the velocities are unequal, then the numerical concentration of the microspheres will be different in different branches. In partic-

ular, if the velocity ratio exceeds the critical value, which for spheres lies in the range 1–2 if the diameters of the spheres approach that of the vessels, then the distribution of the number of spheres will be entirely out of proportion to the ratio of the flows.

These remarks point to a consideration that should be given to the microsphere methods, but its importance depends on the size of the spheres as well as the vascular structure of the specific organs. It also depends on how precise the measurements are intended to be. If the field of view for each measurement is so large that all the variations of concentration of the spheres are averaged out, then our remark has no importance. If the field of view is made smaller and smaller so that a more detailed picture of the concentration distributed is measured, then our remark may become important. By the same token, the larger the spheres, the larger will be the vessels in which the velocities will influence concentration; hence the larger will be the volumes in which local variations of concentration occur to vitiate the simple interpretation of radiation measurements.

Because of the simplicity of the experimental setup, the accuracy of each step is quite good. The velocity measurements are subjected to an error of observing and timing of at most one cell (diam 0.32 cm) in a length of 8.9 cm, or at most 3.6%. The hematocrit measurements are subjected to a trapping error of about 3%, which was corrected, and an observation error of the order of 1%. Hence individual observations are accurate to about 4%. The scatter of the data as shown in the figures must be attributed to the random configurations of the pellets in the flow, geometric, kinematic, and dynamic. After all, particulate flows of the type investigated here are stochastic processes.

APPENDIX

Theoretical Analysis of the Experiment

An approximate analysis of the forces that act on a particle at a bifurcation point will clarify the theme of this paper. We shall show that if the mean velocities of flow in the two daughter branches are $\bar{V}_1$ and $\bar{V}_2$, with $\bar{V}_1 > \bar{V}_2$, then the resultant forces due to pressure and shear stress are both proportional to $\bar{V}_1 - \bar{V}_2$, and act in the direction of the daughter tube with the higher velocity, $\bar{V}_1$. The difference between a rigid spherical particle and a flexible disk will be seen to be significant. The effect of the angle of branching will be shown to be rather minor. Finally, the question of how well the flexibility of the pellets used in the experiment simulates the flexibility of red blood cells in microcirculation is answered.

First, consider disk-shaped flexible pellets. Such a pellet can assume all kinds of orientation in the tube. Let us consider two configurations, one in which the plane of the disk is perpendicular to the axis of the tube, and another in which the plane of the disk is parallel to the axis of the tube. Stated in terms of the angle between the axes of symmetry of the disk and the tube, the angle is zero in the first case and 90° in the second. All other orientations are between these two. For conciseness, we say that the first case is plugging, the second is edge-on.

Consider the plugging case. Since the local velocity of flow must vanish on the wall of the tube and is maximal at the center, the flexible pellet must deform. Consistent with the principle of minimum potential energy, the deformation of a thin disk will be planar, i.e., its midplane will deform into a developable cylindrical surface, which requires no membrane strain in the midplane of the disk. All the energy of deformation is used in bending, and not in stretching the midplane. (If h represents the thickness of the disk, then the bending energy is proportional to h^3 , whereas stretching energy is proportional to h . When $h \rightarrow 0$, it becomes much easier to bend than

to stretch.) Let us sketch such a deformed pellet in a branching tube as shown in Fig. 6, A or B. Let us consider three sections of the tubes as indicated by dotted lines in the figure: one in the mother tube and two in the daughter tubes, all at a distance L from the point of bifurcation. In a flow of a homogeneous viscous fluid before the arrival of the pellet, let the pressures at these sections be P_0 , P_1 , P_2 , respectively. Obviously $P_0 > P_2 > P_1$ on account of the assumption $\bar{V}_1 > \bar{V}_2$. When the pellet arrives at the junction, let the pressures acting on the pellet surface facing the three branches be denoted by P_0' , P_1' , P_2' , respectively, as indicated in the figure. Now, the values of P_0' , P_1' , P_2' depend on how tightly the pellet plugs up the flow. If the flow is momentarily plugged completely, then we have

$$P_0' = P_0 \quad P_1' = P_1, \quad P_2' = P_2 \quad (A1)$$

If the flow can leak past the pellet then $P_0' < P_0$, $P_1' > P_1$, $P_2' > P_2$. But they are related, $P_1' < P_2' < P_0'$; and $P_0' = P_1' = P_2'$ only in the limiting case of no pellet. The force that acts on the pellet that pulls the pellet into branch 1 is equal to the difference of pressures acting on the two sides of the pellet multiplied by the projected area in branch 1; i.e., $(P_0' - P_1')\pi D_c^2/8$. The pressure difference that pulls the pellet into the branch 2 is $(P_0' - P_2')\pi D_c^2/8$. Here D_c is the diameter of the pellet. The resultant force due to the pressure difference is F_p

$$F_p = (P_0' - P_1') \frac{\pi D_c^2}{8} - (P_0' - P_2') \frac{\pi D_c^2}{8} = (P_2' - P_1') \frac{\pi D_c^2}{8} \quad (A2)$$

and is directed into branch 1 because $P_1' < P_2'$. We can convert this expression into the mean velocities of flow as follows. Assuming Poiseuille flow, we have

$$\bar{V}_1 = \frac{D_c^2 (P_0 - P_1)}{32\mu \cdot 2L} \quad \bar{V}_2 = \frac{D_c^2 (P_0 - P_2)}{32\mu \cdot 2L} \quad (A3)$$

where μ is the apparent viscosity of the fluid. If Eq. A1 is applicable (flow plugged momentarily), then a combination of Eqs. A2 and A3 yields

$$F_p = 8\mu L \pi \frac{D_c^2}{D_t^2} (\bar{V}_1 - \bar{V}_2) \quad (A4)$$

In general, Eq. A1 is not applicable, and the best we can say is that

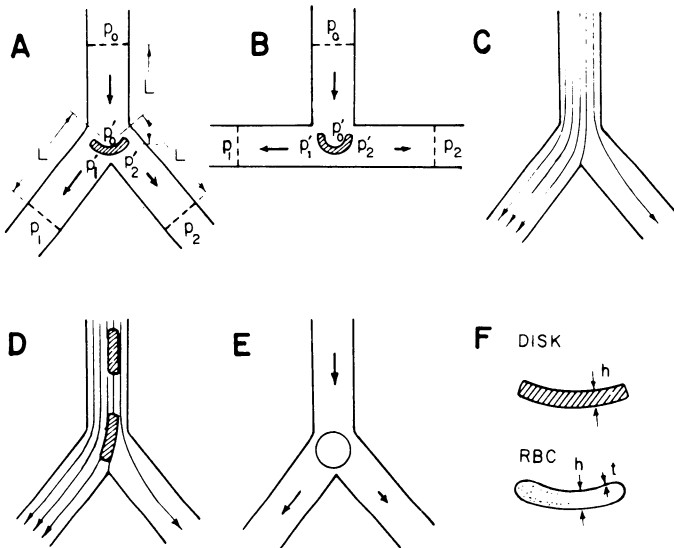


FIG. 6. Illustration of the features of modeling. Flow is directed downward. Velocity in left branch (tube 1) is faster than that in right branch (tube 2). A: a plugging pellet at the fork. B: similar to A, but with branching angle equal to 90° . C: streamline pattern in the case $\bar{V}_1 = 2\bar{V}_2$, as seen in the middle section. D: pellets flowing edge on, and astride of the dividing stream surface. E: flow of a rigid spherical particle at the fork. F: cross sections of a pellet in the model and a red blood cell. Pellet is a solid disk of homogeneous elastic material. Red blood cell is considered to be a thin shell filled with a liquid.

$$F_p = \text{const} \cdot \frac{D_c^2}{D_t^2} (\bar{V}_1 - \bar{V}_2) \quad (A5)$$

This shows that the resultant force is proportional to the velocity difference of the two branches, and to the square of the particle-to-tube diameter ratio.

Next, consider the edge-on case. Here it is necessary to consider the shear stress and velocity gradient, and hence the velocity distribution. The easiest way to understand the velocity distribution in a branching flow is to look at the streamline pattern. A streamline is a line whose tangent is parallel to the local velocity vector. A sketch of the streamline of a flow of a homogeneous fluid in a branching tube is shown in Fig. 6C. We can quantitate the spacing of these lines as follows. Consider an axisymmetric flow in a circular cylinder. The volume rate of flow in a circular ring of radius r and a small thickness Δr is equal to the area of the ring $2\pi r \Delta r$ multiplied by the velocity v at r

$$\Delta Q = 2\pi r \cdot v \cdot \Delta r \quad (A6)$$

We define a stream function $\psi(r)$, which is the value assigned to a streamline at radius r by the requirement that the difference between the stream functions is equal to the flow between the streamlines

$$\Delta Q = \psi(r + \Delta r) - \psi(r) = \Delta \psi = \frac{\partial \psi}{\partial r} \Delta r \quad (A7)$$

A combination of Eqs. A6 and A7 yields

$$v = \frac{1}{2\pi r} \frac{\partial \psi}{\partial r} \quad (A8)$$

Now we can assign $\psi = 0$ at the center and $\psi = Q$ at the tube radius and space the lines $\psi = \text{const}$ in such a way that Eqs. A6–A8 are satisfied. These are shown in Fig. 6C for the case in which $\bar{V}_1 = 2\bar{V}_2$ for the flow of a homogeneous fluid. Note that there is a dividing streamline that stops at the stagnation point of the flow, located at the junction of the daughter tubes. The surface formed by all the dividing streamlines is called the dividing stream surface. All fluid on the left of the dividing stream surface goes into tube 1, all on the right into tube 2.

Now, let there be edge-on pellets in the tube. The edge-on configuration of the pellets will not cause large disturbances in the streamline pattern. If a pellet lies entirely to the left of the dividing stream surface it will flow into tube 1. If it lies to the right, it flows into tube 2. The borderline cases, in which the pellets lie astride the dividing stream surface, as shown in Fig. 6D, are the ones that command attention.

The flow around an edge-on pellet near the bifurcation point can be seen from the streamlines shown in Fig. 6D. The streamlines are crowded on the left side of the pellet, indicating that the flow velocity is high there. On the right the streamlines are rarified. This feature is accentuated especially in the neighborhood of the bifurcation point. It follows that the velocity is higher on the left side of the pellet than that on the right side, because Eq. A8 shows that v is proportional to the density of the streamlines. Since the velocity vanishes on the tube wall, the velocity gradient, i.e., the shear gradient, is higher on the left. Therefore, according to the Newtonian law of viscosity the shear stress acting on the left side of the pellet surface is larger than that on the right. The resultant shear force is obtained by integrating the shear stress over the area. Now the shear gradients on the two sides can be assumed to be proportional to $\bar{V}_1/D_t$ and $\bar{V}_2/D_t$. The resultant shear force is therefore

$$F_s = \text{const} \cdot \frac{D_c^2}{D_t} (\bar{V}_1 - \bar{V}_2) \quad (A9)$$

and is directed toward tube 1.

Pellets in other configurations will be acted on by stress resultants similar to those given by Eqs. A5 and A9, though different in magnitude. Thus, it is seen that the net pressure and shear forces tend to pull pellets to the faster side, more definitely for pellets in the plugging configuration, less for edge-on ones. The magnitude of the force depends on D_c/D_t . The ratio of the number of cells in edge-on configuration to that in plugging configuration varies with D_c/D_t . If $D_c/D_t > 1$, practically all cells will be edge on. The fraction of all edge-on pellets to be astride the dividing stream surface increases as

D_c/D_t increases and as $\bar{V}_1/\bar{V}_2$ increases.

If the pellets were rigid spheres, as shown in Figure 6E, the resultant shear and resultant pressure forces would both help to pull the pellets into tube 1. If $D_c/D_t \rightarrow 1$, the critical velocity ratio of the rigid spheres also tends to 1. The critical velocity ratio increases as D_c/D_t decreases. For flexible disks, it is expected that the critical velocity ratio tends to increase as the pellets become more flexible. These expectations are verified by the experimental results.

Finally, we must answer the question how flexible are the pellets used in our experiments compared with the red blood cells in microcirculation. According to Ref. 8, the gelatin used in our experiments has a shear modulus of 7.7×10^3 dyn/cm² and a Young's modulus of $E = 2.31 \times 10^4$ dyn/cm². On the other hand, the Young's modulus of the red cell membrane is not so well known. The literature has been reviewed by Skalak et al. (5) who estimated the Young's modulus of the red cell membrane to be 10^4 dyn/cm². Hence the elasticity of the gelatin and the red cell membrane are quite similar.

To determine a dimensionless parameter that serves to compare the effect of flexibility of the pellets in the bifurcating flow, we refer again to Fig. 6. The important deformation of the pellet is bending. Suppose a pellet is bent to a curvature $1/R$, R being the radius of curvature. The dimensionless parameter that determines the shape of the pellet is the product of the cell diameter and the curvature: D_c/R . Now for a disk of thickness h , diameter D_c , and Young's modulus E , the curvature is

$$\frac{1}{R} = \frac{\text{bending moment}}{ED_ch^3/12} \quad (A10)$$

The bending moment is proportional to the shear stress $\times$ area $\times$ moment arm, and the shear stress is equal to the coefficient of viscosity of plasma, μ , times the shear gradient. Thus

$$\text{bending moment} = \mu \left(\frac{V}{D_t} \right) \frac{\pi D_c^2}{8} \left(\frac{D_c}{2} \right) \quad (A11)$$

Hence, the flexibility for the model is

$$\left(\frac{D_c}{R} \right)_{\text{model}} = \left(D_c \frac{\pi \mu V D_c^3}{16 D_t} \frac{12}{E D_c h^3} \right)_{\text{model}} = \left(\frac{3 \pi \mu V D_c^3}{4 E D_t h^3} \right)_{\text{model}} \quad (A12)$$

For the red blood cell, when it is considered as a disk-shaped thin shell of uniform thickness h (RBC thickness) with wall thickness t (cell membrane thickness), see Fig. 6F, the curvature is

$$\frac{1}{R} = \frac{\text{bending moment}}{E D_c h^2 t / 4} \quad (A13)$$

Hence the flexibility parameter for the red blood cell is

$$\left(\frac{D_c}{R} \right)_{\text{RBC}} = \left(\frac{\pi \mu V D_c^3}{4 E D_t h^2 t} \right)_{\text{RBC}} \quad (A14)$$

For dynamic similarity between the model and the prototype, we must have

$$\left(\frac{D_c}{R} \right)_{\text{model}} = \left(\frac{D_c}{R} \right)_{\text{RBC}} \quad (A15)$$

i.e.

$$\left(\frac{3 \pi \mu V D_c^3}{4 E D_t h^3} \right)_{\text{model}} = \left(\frac{\pi \mu V D_c^3}{4 E D_t h^2 t} \right)_{\text{RBC}} \quad (A16)$$

Note that the velocity of flow enters into this equation. Typical values of our experimental model are $\mu = 100$ poise, $V = 1$ cm/s, $D_c = 3.2$ mm, $h = 1$ mm, $D_t = 4$ mm, $E = 2.31 \times 10^4$ dyn/cm². Typical values for RBC in capillaries are $\mu = 0.01$ poise, $V = 0.1$ cm/s, $D_c = 8$ μ m, $h = 1.5$ μ m, $D_t = 10$ μ m, $t = 50$ Å, $E = 10^4$ dyn/cm². Hence, typically,

$$\left(\frac{D_c}{R} \right)_{\text{RBC}} = 3.6 \text{ (V in mm/s)} \quad (A17)$$

$$\left(\frac{D_c}{R} \right)_{\text{model}} = 0.8 \text{ (V in cm/s)} \quad (A18)$$

Thus, to simulate a capillary blood flow at 1 mm/s the flow velocity in the model should be 4.3 cm/s.

The simulation of the flexibility parameter of the red blood cell is quite good in our model according to our present understanding of the red cell membrane.

The theory presented above is simple because many details are left in a qualitative state. For future refinement, the probability of finding red cells in the edge-on and plugging or other configurations must be analyzed, and the resultant pressure and shear forces must be more accurately assessed.

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